

Prophase I:

It consists of the longest phase of meiosis. It can be subdivided into following sub stages:

- | | | |
|---------------|----------------|---------------|
| (1) Leptotene | (2) Zygotene | (3) Pachytene |
| (4) Diplotene | (5) Diakinesis | |

(1) Leptotene:

During this sub stage following changes occur; The chromatin network break into specific number of long thin beaded thread called leptotene. Each thread has two morphologically similar leptene in each cell called homologous structure.

(2) Zygotene:

During this sub stage the Homologous (Similar structure) chromosomes, which comes from the mother (by ovum) and father (by sperm) are attracted towards each other and their lengthwise pairing takes place. The pairing of homologous chromosomes is known as **synapsis**, while the paired homologous chromosomes are known as **bivalent**.

(3) Pachytene:

The synaptic forces of attraction between each bivalent decrease and the chromosomes uncoil and separate. The separation is however incomplete and paired chromosomes are in contact with each other at one or more points, called Chiasmata. Each homologous chromosome split longitudinally except in the centromere region. Now each bivalent is composed of four chromatids and therefore is known as **bivalent tetrad**.

(4) Diplotene:

The homologous chromosomes exchange their parts of chromatid at Chiasmata. This exchange of segments of chromatids at chiasmata between the homologous chromosomes is called Crossing Over.

(5) Diakinesis:

During this sub stage; nucleoli and nuclear membrane are disappeared, whereas Mitotic Apparatus (spindle) is completed. Chiasmata moves from the centromere towards the ends of the chromosomes like a zipper.

This type of movement of chiasmata is known as Terminalization. At the end of Diakinesis chromatids still remain compacted at their ends.

Metaphase I:

Following changes occur in this phase:

The bivalent line up at the equatorial plane. The centromere of each chromosome attaches with same fibres of spindle.

Anaphase I:

At this stage one chromosome from each member of homologous pair (bivalent) begins to separate and move towards its respective pole by the contraction of spindle fibers.

The actual reduction occurs at this stage because half the number of chromosomes moves to each pole. Moreover as a result of crossing over the two chromatids of a chromosome do not resemble with each other in the genetic terms.

Telophase I:

The nuclear membrane form around the chromosomes at each pole and chromosomes become uncoil. The nucleolus reappears and thus two daughter nuclei formed.

Cytokinesis: Telophase may or may not be accompanied by cytokinesis and daughter cells formation.

Interphase: Following Telophase I (If this stage occurs), there is a short period called **Interphase** before meiosis II. It is similar to Interphase between Mitotic divisions except that DNA replication does not occur. Replication of DNA is unnecessary because each chromosome already has two chromatids.

Second Meiotic Division (Meiosis II):

The second meiotic division is actually the mitotic division which divides each haploid cell formed during meiosis I into two daughter haploid cells. The second meiotic division includes:

- (1) Prophase II (2) Metaphase II (3) Anaphase II (4) Telophase II

(1) Prophase II:

The spindle fibres are formed. The nuclear membrane and the nucleolus disappear.

(2) Metaphase II:

The chromosomes are attached to half spindle fibers by their centromere and get arranged at equatorial plane. Each chromosome attach with separate fiber of spindle.

(3) Anaphase II:

The spindle fibers attached to the centromeres shorten and the chromatids of chromosomes are pulled away from one another. This movement continues until one complete set of chromosomes is positioned at each pole of the cell.

(4) Telophase II:

The spindle fibers disappear completely and chromosomes begin to uncoil. They become longer and indistinct and form group at each pole. Around each group nuclear envelope is formed.

After the karyokinesis in each haploid meiotic cell the cytokinesis i.e. division of cytoplasm occurs and thus four haploid cells are formed.

What happens in absence of meiosis?

In the absence of meiosis number of chromosomes would have been doubled giving rise to abnormal growth, changes in species characteristics and or may prove fatal.

Significance of Meiosis:**(1) Constant number of chromosomes:**

Meiosis maintains chromosome number constant from generation to generation. It is due to the fact that meiosis reduces the diploid number of chromosomes to half i.e. haploid in the gametes.

During fertilization the diploid number of the chromosomes is restored.

(2) Responsible for genetic variation among species:

By crossing over, the meiosis provides an opportunity for the exchange of the genes between homologous chromosome and thus cause the genetic variations among the species. The variations are the raw material of the evolutionary process.

Meiotic errors:

In normal course of meiosis, the two chromosomes of each homologous pair separates and enter into two gametes, but sometimes a pair of homologous chromosomes fails to separate from one another, during meiosis I. this phenomenon is called Non-disjunction.

Non-disjunction produce gametes with abnormal number of chromosome i.e either with less or extra chromosome. If such abnormal gametes fuse with normal gametes, the resulting zygote will also have abnormal number of chromosome.

Summary

- Term chromosome is given by *Fleming* in 1882. These are thread like structures appear at the time of cell- division made up of chromatin material in specific number in Eukaryotic cell.
- Chromosomes are made up of DNA and histone protein.
- There are four types of Chromosome i.e Metacentric, Sub-metacentric, Acrocentric and Telocentric.
- Sequences of change occur between two successive Cell division called cell-cycle.
- Cell-cycle is mainly consist of two phases i.e cell-division and inter phase.
- Inter phase is divided further into three sub phases G₁, S and G₂ phases.
- Mitosis is the type of cell division, where a parent cell divides into two daughter cells, with same number of chromosomes as the parent cell contain.
- Meiosis is the type of cell division in which a single cell divides into four daughter cells and number of chromosomes become half of the parent cell.
- In animal meiosis takes place in the Germ cells while in plant it takes spore mother cells to produce gametes and spores respectively.
- Meiotic error, where a pair of homologous Chromosomes fails to separate from one another called non disjunction. This non disjunction produce gametes with abnormal number of chromosomes with less or extra chromosomes.

- Cell-death occur in two ways:
 - (i) Apoptosis; programmed cell death, which help in proper control of development.
 - (ii) Necrosis; cell death occur by external factor or accidental cell-death.

Review Questions

1. Encircle the correct answer:

- (i) Which processes involve mitosis?
 - (a) Growth, reduction division and asexual reproduction.
 - (b) Growth, repair and asexual reproduction.
 - (c) Growth, repair and semiconservative replication.
 - (d) Growth, reduction division and repair
- (ii) What occurs in metaphase of mitosis?
 - (a) Chromosome line up on the equator of the cell.
 - (b) Chromatids reach the poles of the spindle.
 - (c) Chromatids separate and move to opposite poles
 - (d) Chromosomes start to coil up and become visible.
- (iii) Select the mismatched
 - (a) Anaphase → Chromatids move
 - (b) Prophase → Centriole move
 - (c) Telophase → Nuclear membrane disappear
 - (d) Metaphase → Chromosome line up.
- (iv) Which process occurs during prophase of mitosis in an animal cell?
 - (a) Division of centromeres
 - (b) Formation of chromosomes
 - (c) Replication of DNA
 - (d) Separation of centrioles
- (v) Sequence of physiological changes by which cell commit suicide.
 - (a) Apoptosis
 - (b) Necrosis
 - (c) Autophagy
 - (d) Both "a" and "b"

- (vi) Choose the incorrect statement regarding meiosis.
- (a) Maintain chromosome number from generation to generation.
 - (b) Reduces the diploid number of chromosomes to half
 - (c) Takes place in germ cell to produce gametes
 - (d) Production of new somatic cells from germ cells.
- (vii) A type of cell division in which spore mother cell produce spores.
- (a) Amitosis
 - (b) Mitosis
 - (c) Meiosis
 - (d) Both "a" and "c"
- (viii) Stage of mitosis in which chromatids reach the poles and their movement ceases.
- (a) Prophase
 - (b) Metaphase
 - (c) Anaphase
 - (d) Telophase
- (ix) Stage of meiosis in which centromeres shorten and the paired chromatids are pulled away from one another.
- (a) Anaphase-II
 - (b) Metaphase-II
 - (c) Telophase-II
 - (d) Prophase-II
- (x) Phenomenon in which pair of homologous chromosome fails to separate called.
- (a) Non-disjunction
 - (b) Terminalization
 - (c) Synapsis
 - (d) Linkage

2. Fill in the blanks:

- (i) Chromosomes are thread like structures appear at the time of _____.
- (ii) Sequence of changes which occurs between one cell division and the next is called _____.
- (iii) The two chromatids are attached to each other at _____.
- (iv) Chromosomes in the cell which are similar to each other in shape and size are called _____.

- (v) Chromosomes with one arm very small and other very long called_____.
- (vi) Chromosomes consist of two genetically identical copies of thread called_____.
- (vii) Period of extensive metabolic activity, in which cell grow, enzymes are synthesizes_____.
- (viii) In animal meiosis produces_____.
- (ix) During metaphase homologous chromosomes arrange themselves on the _____.
- (x) Cell death which is caused by external factor called_____.

3. Define the following terms:

- | | | |
|-----------------------|------------------|--------------------|
| (i) Pachytene | (ii) Cytokinesis | (iii) Bivalent |
| (iv) Chiasmata | (v) Chromatids | (vi) Diakinesis |
| (vii) Terminalization | (viii) Necrosis | (ix) Crossing over |
| (x) Centromere | | |

4. Distinguish between the following in tabulated form:

- | | |
|------------------------------|-----------------------------|
| (i) Prophase and Prophase-I | (ii) Prophase and Telophase |
| (iii) Apoptosis and Necrosis | |

5. Write short answers of following questions:

- (i) Why meiosis-I is called reduction division?
- (ii) Why mitosis is necessary for growth?
- (iii) How number of chromosomes remains constant from generation to generation?
- (iv) Why interphase is called as phase of high metabolic activities?
- (v) Why interphase between meiosis-I and meiosis-II is short?

6. Write detailed answers of the following questions:

- (i) Describe various stages of mitosis with suitable diagrams.
- (ii) Describe stages of meiosis-I with suitable diagrams.

ENZYMES

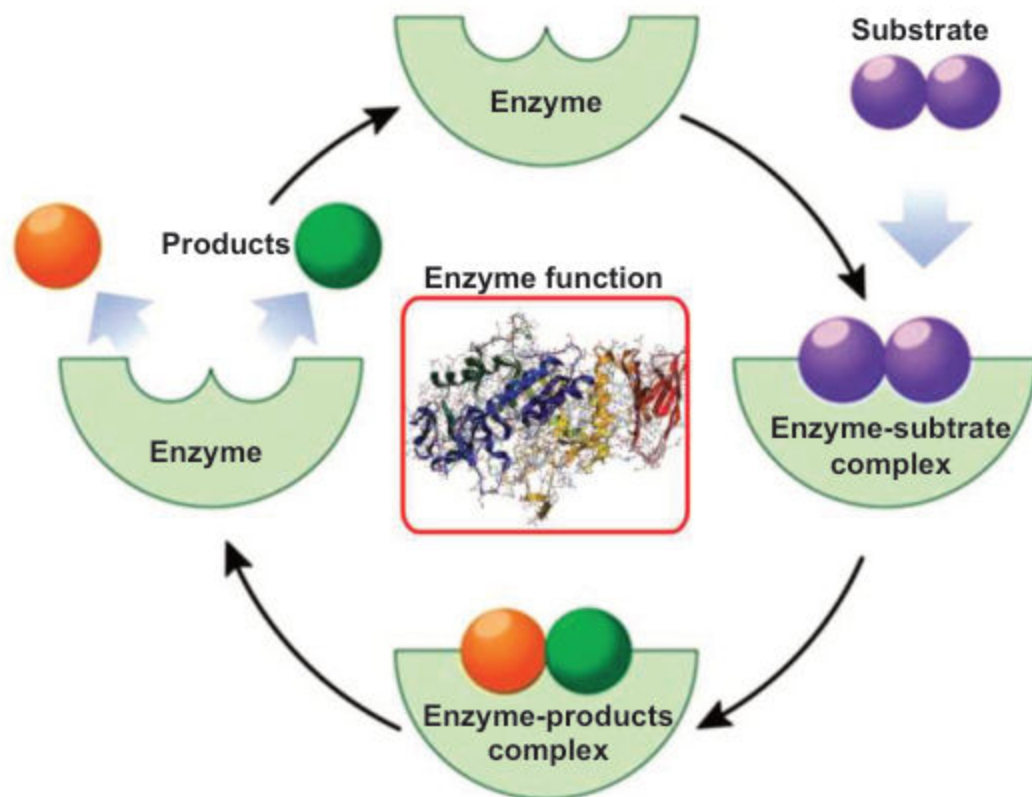
Chapter

6

Major Concept

In this Unit you will learn:

- Definition and Characteristics of Enzymes
- Mechanism of Enzyme Action (Lock-n-key Model)
- Specificity of Enzyme



Life is another name of activity, therefore thousands of chemical reactions take place in the body of an organism. These reactions of an organism are collectively called metabolic reactions and this phenomenon of chemical activity called metabolism. The metabolic activities are always of two types, either constructive or destructive.

In constructive reactions large molecules are formed to form a structure of cell or body. These reactions are called anabolic reactions and this type of metabolism is called **anabolism**. On the contrary, the destructive reaction in which large molecules breakdown in small molecules to produce energy or to re-utilize further or to discard called catabolic reactions. The type of this metabolic activity is called **catabolism**.

The chemical reaction requires particular conditions to carry down at proper rate, especially temperature and pressure. The conditions of temperature and pressure inside cell or organism are generally found not suitable for chemical reactions e.g. inside human body normal temperature remain 37°C and pressure is 120/80 m.m of Hg. These conditions of temperature and pressure are not enough to perform any chemical reactions. Now question arises here how biochemical or metabolic reactions can occur without altering these conditions?

Now body requires some facilitators. These facilitators helps to perform biochemical reactions at low energy. From above discussion it is clear now that each reaction requires some amount of minimum energy to initiate a reaction. This minimum amount of energy is called **activation energy**. If this amount is high the difficult will be the reaction or vice versa e.g. the activation energy needed to break a glucose molecule initially requires energy of 2 ATP molecules.

6.1 DEFINITION

The high amount of activation energy cannot be provided by organism itself therefore they require some facilitators to reduce this activation energy. These facilitators are special molecules made up of mostly protein called **enzymes** (En=inside, zyme = yeast). The name was coined due to observation when yeast was introduced in fruit sap which converted it into alcohol. Now the enzyme are defined as the biocatalyst which facilitate chemical reaction by lowering activation energy.

This action of enzyme allows biological reaction to proceed rapidly at relatively low temperature and pressure tolerable by living organism.

6.2 CHARACTERISTICS OF ENZYMES

- Enzymes are biocatalyst, made up of mostly proteins and therefore three dimensionally folded chains of amino acids with a specific shape. This shape is determined by the sequence of amino acids held together by bonds, for example Hydrogen bonds. Enzymes speed up reactions by bringing reactants together and reducing the activation energy required to start the reaction (enzymatic reaction)
- When an enzyme starts a chemical reaction, catalyzes the reaction hence does not utilized itself which means even a single or little amount of enzyme can start a reaction and catalyze fastly.
- Their presence does not affect the nature or properties of end products.
- Reactants of enzyme are called substrate.
- They are very specific in their action; a single enzyme catalyzes only a single chemical reaction or a group of related reactions.
- A small portion of enzyme where substrate attaches with enzyme is called **active site**. The shape of active site is complementary to shape of the substrate.
- They are sensitive to even a minor change in pH, temperature and substrate concentration.
- Some enzymes require cofactor for their functioning; a cofactor is a non-protein substance which may be organic or inorganic. Zn^{+2} , Mg^{+2} , Mn^{+2} , Fe^{+2} , Cu^{+2} , K^{+1} and Na^{+1} the organic cofactors are NADP, NAD and FAD are used in enzymes as cofactors.

Cofactor can be categorized into prosthetic group (if organic cofactors are tightly bound to an enzyme) and Coenzymes (if organic cofactors are loosely attached with an enzyme)

- Many enzymes work in a sequential manner to produce a specific product. This pathway is called metabolic pathway.
- Activity of enzymes can be enhanced by activator and can be decreased by inhibitors.
- An **enzyme inhibitor** is a molecule that binds to an **enzyme** and decreases its activity. Since blocking an **enzyme's** activity can kill a pathogen.

6.2.1 Uses of enzymes:

Many enzymes are used commercially in industries. The most common industries are:

- **Paper industry-** To get cellulose for paper making.
- **Food industry-** For making bakery products and pizza.
- **Brewing industry-** For conversion of sugar into alcohol.
- **Bio-detergents-** Use to remove different type of stains.

6.2.2 Factors affecting the activity of an enzymes:

In nature, organisms adjust the conditions of their enzymes to produce an optimum rate of reaction, where necessary, or they may have enzymes which are adopted to function well in extreme conditions where they live.

Substrate Concentration:

It has been shown experimentally that if the amount of the enzyme is kept constant and the substrate concentration is then gradually increased, the reaction velocity will increase until it reaches a maximum after which further increase in the substrate concentration produces no significant change in the reaction rate.

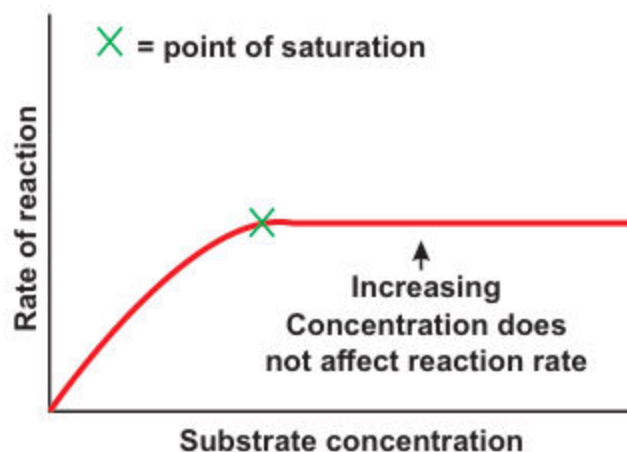


Figure 6.1 Effect of substrate concentration on enzyme activity

In other words, the enzyme molecules are saturated with substrate. The excess substrate molecules cannot react until the substrate already bound to the enzymes has reacted and been released (or been released without reacting).

Temperature:

The protein nature of the enzymes makes them extremely sensitive to thermal changes. Enzyme activity occurs within a narrow range of temperatures compared to ordinary chemical reactions. Enzymes catalyse by randomly colliding with substrate molecules, increasing temperature and increases collision which also increases the rate of reaction,